

Land Transport

Authority:

Collecting travel data securely at

scale and with ease to

improve

transport

planning with Google Cloud

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

[About Land Transport](#)



About Land Transport Authority

Land Transport A
launches its digit
survey on Google
Kubernetes Engin
connect and engi

Established in 1995, the Land Transport Authority (LTA), is a statutory board in charge of planning, designing, constructing, and maintaining Singapore's transportation infrastructure and systems. Using technology to enhance the reliability of the public transport system in Singapore, its mission is to connect people and places and enhance the travel experience in the country.

commuters, while
leveraging Looker
for easy reporting

Industries: Transportation,
Government & Public Sector

Location: Singapore

Google Cloud results

- Increases marketing effectiveness by analyzing survey performance in Looker Studio to identify underrepresented groups of commuters
- Ensures data security and protect against vulnerabilities with Cloud IAM and Cloud Armor
- Improves data quality with fraud detection logic to detect bot traffic with Cloud Monitoring
- Reduces survey cost from \$6.5 million to

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(<https://cloud.google.com/functions/>), Cloud



About Affle

Affle, founded in 2005, is a global technology company that uses its consumer intelligence marketing platform to help businesses increase user acquisitions, engagements, and transactions. Affle has received numerous accolades, including Technology Company of the Year at the MMA Smarties 2020. As a Google Cloud partner, Affle provides enterprise businesses with a variety of solutions, including platform-based app development.

Google Cloud (<https://cloud.google.com/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Google Cloud Armor
(<https://cloud.google.com/armor/>)

Cloud Build (<https://cloud.google.com/build/>)

Affle, founded in 2005, is a global
technology company that uses its

approximately
\$2.8 million with
Google Cloud

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Operations

(<https://cloud.google.com/products/operations>)

Monitoring (<https://cloud.google.com/monitoring/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Every day, millions of commuters
on the Land Transport Authority

(<https://www.lta.gov.sg/>) (LTA)'s
transport network to get to school

the city. This is why it is crucial to
reliable and efficient transportation
the current and future needs of

With a vision to build a people-centric
system, LTA launched its digital

Survey

(https://www.lta.gov.sg/content/ltagov/work/lta_surveys.html)

on Google Cloud (<https://cloud.google.com/>) in 2021 to

understand travel patterns and provide insights for
future land transport planning.

“As a result of
our digital
survey platform
on Google Cloud,
we were able to
collect
responses

during the
COVID-19
pandemic
without in-
person contact.

We can adjust
our marketing
approaches by
tracking real-
time survey
results on
Looker Studio.

This is a key
accomplishment
as we need to
ensure that we
collect
responses from
all segments of
commuters and
not just the
young or the
digitally savvy.”

—Baskaran Narayanan,
Senior Manager,
Transport Planning Lab,
Strategic Planning, Policy
& Planning Group, Land
Transport Authority

“We launched the household travel survey to better understand how commuters use our transport systems, and how we can use these insights for future land transport planning,” says Baskaran Narayanan, Senior Manager, Transport Planning Lab, Strategic Planning, Policy and Planning Group at LTA. Starting with a six-week pilot survey in April 2021, the household travel survey will continue to run through to 2025 to reach approximately 52,000 responses.

Previously, LTA conducted in-person door-to-door surveys every four to five years. In a bid to streamline work processes and collect survey responses more efficiently, LTA embarked on the journey to digitalize the survey through the use of new technologies. “As a result of our digital survey platform on Google Cloud, we were able to collect responses during the COVID-19 pandemic without in-person contact. We can adjust our marketing approaches by tracking real-time survey results on Looker Studio. This is a key accomplishment as we need to ensure that we collect responses from all segments of commuters and not just the young or the digitally savvy,” says Baskaran.

By hosting a digital survey on Google Cloud, LTA can promote the survey across multiple websites, allowing respondents to complete the survey at leisure. Already, the team is seeing an improvement in response rates compared to before, with the

digital survey showing seven times higher responses than that of manual surveys. In addition to the increase in efficiency in gathering data, going digital has also resulted in time and cost savings.

The survey platform was run on a fully managed Google Kubernetes Engine (/kubernetes-engine) (GKE) which allowed the team to focus on monitoring survey responses, enhancing user experience, and managing user feedback. LTA uses a combination of Cloud SQL (/sql) and Cloud Storage (/storage) to store more than one million responses and creative assets securely. It also uses Looker Studio (<https://marketingplatform.google.com/about/data-studio/>) to visualize data and share reports with relevant agencies.

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Collecting survey responses at scale and speed with GKE

Affle helped LTA with all stages of the survey, from design to data analysis. It is the project team's primary objective to build a user-friendly interface that is easy to understand, so respondents can fill out the survey with ease. To improve user experience, Affle performed beta testing to spot bugs and detect any unresponsive gestures and navigation issues. Setting up a CI/CD pipeline with Cloud Build (/cloud-build) enables the project team to

test, build, and deploy new features such as the visibility of language options and notification functions. With automated deployment for GKE, routine DevOps tasks such as starting up new services and configuring staging environments can be done with a few clicks. New features and fixes can be deployed more quickly and reliably, contributing to a frictionless user experience.

In contrast to manual surveys, where the surveyor only conducts one survey for each household they visit, online marketing tactics can increase the response rate for online surveys. Each time a marketing campaign is launched on a new partner site, an increase in responses is seen in a shorter period of time.

“It is important to balance platform performance and cloud costs. While survey responses tend to come in large waves, it doesn’t make sense for us to maintain a large number of VMs just for the possibility of traffic peaks. Using Google Kubernetes Engine, Affle and Google Cloud designed a seamless user experience that scales dynamically as traffic increases,” says Baskaran.

Affle uses Cloud Functions (/functions) with GKE to run auto-scaling instead of over-provisioning resources. At the same time, Cloud Monitoring (/monitoring) provides real-time activity monitoring, so that the organization doesn’t end up wasting resources on fraudulent traffic such as spam or bots.



Building user trust in data collection with Google Cloud IAM

Based on previous data, LTA found that people are most concerned about the security of their personal information like telephone numbers and household income when filling out the forms. This is why data privacy is a key priority for LTA. Unlike manual household travel surveys, in which a third party collects and enters survey responses, only authorized staff at LTA can access the digital household travel survey data.

“Since the Household Travel Survey is a public-facing government application, it must follow strict security and data privacy standards,” says John Ng, Senior Manager, Data Science at LTA. “In order to secure our survey data, we use Identity and Access Management (/iam) (IAM) to assign user rights to specific resources and create an audit trail to track all activity. Using predefined rules, Cloud Armor (/armor) protects our platform against common vulnerabilities.”

LTA also needs to set parameters that filter out bot responses by hackers who may attempt to cheat the system for the survey incentive. By using mFaaS (<https://mfaas.com/>) machine learning algorithms, Affle flags suspicious responses so LTA can review the surveys for before the incentive is approved.

“Affle’s knowledge of Google Cloud technologies helps us build, deploy, and secure the codebase very smoothly,” says Baskaran. “Affle’s mFaaS solution also helps to protect our survey from abuse so we’re not paying incentive for fraudulent responses.”

Boosting data insights with visualization on Looker Studio

The results of LTA's household travel survey are crucial to several groups of government stakeholders. LTA's strategic planning team analyzes statistics to develop long-term travel plans. Using proprietary software, the transport modeling team forecasts travel demand based on different housing profiles. Data is also used by LTA to study and adjust policies such as public transportation infrastructure and planning for future land transport networks.

In the past, LTA relied on market research personnel to manually enter responses into spreadsheets. This method is time consuming and hampers real-time analysis because the database only reflects point-in-time data. Now, LTA uses Looker Studio to track real-time metrics such as survey completion across location, time, and demographic backgrounds to target groups with lower response rates.

“Using Looker Studio makes reporting a breeze.

Data from multiple sources can be blended and visualized in a single report that can be shared easily,” says Rachel Lim, Data Scientist at LTA.

“Viewers can easily select the period to analyze with the help of date range filters and interactive tooltips.”

In Looker Studio, LTA uses different charts to present survey data effectively. Time series line charts allow viewers to compare survey completion rates over time. Bar charts and pie charts allow viewers to identify trends between groups and develop targeted strategies in response. For reporting purposes, admin users can download the charts or view them on an interactive dashboard.

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patterns over
time to forecast
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and make
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Accelerating a sustainable transport system that meets long- term needs

LTA is driven by the vision of a people-centric land transport network that connects communities and places. The company aspires to strengthen

Singapore's land transport connectivity and integrate a greener and more inclusive public transport system, complemented by walk and cycle options. This digital household travel survey is a first, and allows LTA to harness new technology and innovation to continually meet the evolving travel needs of Singapore's diverse commuters.

"Google Cloud has significantly enhanced how we collect and handle survey data. We would like to explore the potential of its machine learning capabilities to boost our data initiatives, extracting previously unknown patterns over time to forecast future demand and make confident planning decisions for a sustainable and reliable transportation system," says Baskaran.